04 Ch 04 – Security Analysis

Asked in last 3 papers: all theory, 4m each – technical analysis arguments for/against (Sep 2025), identify chart patterns (Jan 2026), Elliott Wave Theory (May 2026).

Introduction

→ **Buying** a security: highest return per unit of risk or lowest risk per unit of return. **Selling** needs no such test. Attractiveness changes with management policy or government economic policy → analyse **continuously**.

→ **Security analysis**: systematic analysis of risk-return profiles so a rational investor can estimate a value from all price-sensitive information and buy when the market **under-prices**.

→ Two approaches: **fundamental** (factors affecting risk-return) and **technical** (demand/supply of securities and share price trends).

1 Fundamental Analysis

→ Assumes share price depends on **future dividends** expected. PV of future dividends at an appropriate discount rate = **intrinsic value** (true value).

→ Price **below** intrinsic value → **buy**; price **above** → **sell**.

Formula	Terms / logic
$P_0 = D/(1+k) + D/(1+k)^2 + D/(1+k)^3 + \ldots$ (eq. 2); $P_0 = D \div k$ (eq. 3)	Uniform dividend D every year; k = desired return on share. Perpetuity.
$P_0 = D_1 \div (k - g)$ (eq. 5); $P_0 = D_0(1+g) \div (k - g)$ (eq. 6)	DPS grows at g p.a. (perpetual geometric series). $D_1 \approx D_0(1+g)$. Valid only if **g < k**; g ≥ k gives infinite price (not restrictive — long-term dividend growth unlikely to exceed k). **Gordon's dividend growth model** (Myron J. Gordon). g = 0 → back to eq. 3.

1.1 Dividend Growth Model and the PE Multiple

Formula	Terms / logic
$P_0 = bE_1 \div (k - g)$ (eq. 7); $P_0 = bE_0(1+g) \div (k - g)$ (eq. 8)	b = dividend payout ratio, so $D_1 = bE_1$; E = EPS. Used to estimate the fundamental price or **fundamental P/E** multiple.

→ Prevailing price / P/E **higher** than fundamental value → overpriced → **sell** (price falls once information is common knowledge). **Lower** → underpriced → **buy**.

→ Industries are interlinked and tied to the economy; a company depends on its industry and its competitive position in it → fundamental analysis = **economy** → **industry** → **company** analysis (three concentric circles = stages of the investment decision).

1.2 Economic Analysis

→ Assess macro-economic factors: past/present performance and future expectations, sector growth, signs of stagnation. Trends in people's **income and expenditure** reflect future industry growth; consumption affects profits, dividends and share prices.

1.2.1 Factors Affecting Economic Analysis

- → **Growth rates of national income**: nominal GDP growth vs **real** growth (after inflation); pointer to returns from shares.
- → **Growth rates of industrial sector**: by industry, estimated from demand for products.
- → **Inflation**: WPI or CPI; affects demand esp. consumer products → firms fine-tune pricing, distribution, promotion.
- → **Monsoon**: strong forward and backward linkages.

1.2.2 Techniques Used in Economic Analysis TYK T4

Forecast national income and components; **GNP** measures national income and is a forecasting tool.

- → **(a) Anticipatory surveys**: expert opinion on construction, plant & machinery spending, inventory levels, consumer spending habits. Drawbacks: (i) intentions may not materialise; (ii) not forecasts per se (consensus approach). Needs continuous monitoring.
- → **(b) Barometer / indicator approach**:
  - → **Leading** — reach high/low points **in advance** of economic activity.
  - → **Roughly coincidental** — peaks and troughs at about the **same time** as the economy.
  - → **Lagging** — turning points **after** the economy.
  - → Limitation: show **direction, not magnitude**; may give conflicting signals → use a **diffusion / composite index** (several indicators in one); difficult to compute and does not eliminate irregular movements.
  - → **Money supply**: affects GNP, profits, interest rates, stock prices; more money supply fuels inflation; stocks are a hedge against inflation → prices rise in inflation.
- → **(c) Economic model building** (GNP model / sectoral analysis, national accounting framework). Steps: (i) hypothesise total demand via GNP (political stability, inflation, economic levels); (ii) forecast components — **consumption, gross private domestic investment, government purchases, net exports**; (iii) add up to forecast GNP; (iv) compare with an **independent agency's** forecast and test for consistency.

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TYK T4 (indicators to assess performance of an economy) → ICAI answer: refer para 1.2.2.

1.3 Industry Analysis

Industries do not grow at the same rate → examine demand, cost structure, economic and government constraints; company profitability depends on its industry's prospects.

1.3.1 Factors Affecting Industry Analysis

- → **Product life-cycle**: high profitability in initial and growth stages, medium but steady in maturity, sharp decline in last stage.
- → **Demand-supply gap**: excess supply cuts unit price realisation and profit; insufficient supply improves it.
- → **Barriers to entry**: high profits attract entrants; barriers innate to product/technology or created by existing firms.
- → **Government attitude**: crucial determinant.
- → **State of competition**: leaders and nature of competition (foreign/domestic), homogeneous vs differentiated products, demand by customer/area, changes in demand pattern, type of industry — **growth, cyclical, defensive, decline**.
- → **Cost conditions and profitability**: (i) cost allocation (material, labour, overheads) and controllability; labour cost (wages, productivity); (ii) product price; (iii) capacity — installed, idle, operating; (iv) capex for maintenance/efficiency. Ratios: G.P., operating profit margin, ROE, return on total capital.
- → **Technology and research**: (i) nature of technology; (ii) expected changes → new products, sales; (iii) capex vs sales over time; (iv) R&D spend — redundant or not; (v) sales and profitability in short, immediate, long run.

1.3.2 Techniques Used in Industry Analysis

→ **Regression analysis**: product demand analysis — GNP, disposable income, per capita consumption/income, price elasticity; regression and correlation.

→ **Input-output analysis**: flow of goods/services through the economy from raw material to final consumption → detect growth/decline of industries.

1.4 Company Analysis

Examine the company's quantitative and qualitative fundamentals.

→ **(a) Net worth and book value**: NW = equity + preference capital + free reserves − intangibles − carried-forward losses; ÷ shares = book value. Not intrinsic worth because (1) market price reflects **future earnings**, not assets; (2) book value is **historical cost**, may underestimate replacement/resale value.

→ **(b) Sources and uses of funds** (funds flow analysis): check if **short-term sources fund long-term investments** → liquidity crunch risk (maturity mismatch).

→ **(c) Cross-sectional and time series analysis**: compare firms, against industry benchmarks, and over time — **common-size statements, ratio analysis**.

→ **(d) Size and ranking**: net capital employed, net profit, ROI, sales vs peers; technical know-how, R&D, price leadership → risk.

→ **(e) Growth record**: growth in sales, net income, capital employed, EPS. Three indicators: **P/E ratio, % growth of EPS p.a., % growth of net block**. Net block growth shows capacity build-up/diversification. Assess capacity utilisation (P&L quantitative info), expansion plans (Directors' Report, Chairman's statement, capital commitments), technology. **Growth is the single most important factor** in company analysis.

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> **SM example (P/E):** A — price 150, EPS 25 → P/E **6**; B — price 250, EPS 25 → P/E **10**. Other factors similar → A preferable. EPS cannot be the sole basis, but P/E relates market price to EPS.

Note (app-written): the SM then says to ascertain "why the EPS in company B is comparatively low", but both EPS are ₹ 25 in the SM table; it is B's P/E that is higher.

→ **(f) Financial analysis**: solvency, liquidity, fund-use efficiency, profitability, operating efficiency, financial and operating leverage. Key five: **EPS, P/E, yield, book value, intrinsic value** — for ~10 years, vs peers and industry averages. Yield and asset backing decide if price is proper. Solvency: current ratio, liquidity ratio, debt-equity.

→ **(g) Competitive advantage**: drives long-term success; shield against competitors (SM: Apple's brand, Samsung's mobile domination).

→ **(h) Quality of management**: intangible; business houses command premium; experience, skills, integrity; dividend/bonus policy. Small investors can't meet top executives. Clues: **conference calls** (CEO/CFO Q&A), **MD&A report**, management holding shares (large unloading while saying otherwise = warning), past 5-year performance of executives.

→ **(i) Corporate governance**: compliance with **SEBI (LODR) Regulations 2015**; how policies serve stakeholders; quality and timeliness of disclosures; quality independent directors.

→ **(j) Regulation**: Companies Act, SCRA, SEBI Act; SEBI monitors listed companies; FEMA compliance for dealings outside India, monitored by RBI.

→ **(k) Location and labour-management relations**: power, skilled labour, raw materials, nearness to markets; state of labour relations.

→ **(l) Pattern of existing stock holding**: stakes of parties (SM case: PNB nationalisation — institutional investors' bargaining raised compensation for opting-out shareholders).

→ **(m) Marketability**: listing ≠ liquidity; see dispersal / public holding, speculative interest, exchange, trading volume.

Techniques used in company analysis *(statistical):*

→ ***Correlation and regression****: simple (2 variables) / multiple (more); links economy, industry, company variables; gives forecasts and tests reliability.*

→ ***Trend analysis****: one variable over time using regression; historical behaviour.*

→ ***Decision tree analysis****: range of values with probabilities; sequential decisions — multiply probabilities along each sequence, then sum. Simulation also reduces limitations.*

→ *Fundamental analysis = examination of economic and financial aspects to estimate* ***future earnings and dividends****, incl. macro-economic and political factors → buy/sell decision.*

→ ***Group analysis****: SEBI wants disclosure of group companies' financial health and promise vs performance in offer documents; RBI focuses on* ***Group Exposure Norms*** *of banks.*

2 Technical Analysis · 2.1 Meaning

→ *Study of share price movements through* ***price graphs/charts****, assuming price trends are* ***repetitive*** *because investor psychology follows a pattern. Studies* ***investor and price behaviour****.*

→ *Two questions: (i) is there a discernible* ***trend****? (ii) are there indications it will* ***reverse****? Methods:* ***visual*** *(charts, patterns) and* ***statistical*** *(price and return data).*

Note (app-written): the SM prints "The technical analyst is concerned with the fundamental strength or weakness of a company"; the rest of the SM (2.2(iv), section 3) shows technical analysis uses market data, not fundamentals — read as "not concerned".

2.2 Assumptions

→ *Market value depends on* ***supply and demand*** *for a security.*

→ *Supply and demand governed by factors* ***rational or irrational*** *(e.g. Govt NPA initiatives → demand for bank stocks).*

→ *Prices move in* ***trends*** *that continue for a substantial period (in a bull market a substantial correction gives a buying opportunity).*

→ *Relies on* ***chart analysis*** *of past prices, not financial statements.*

2.3 Principles of Technical Analysis

→ ***Market discounts everything****: price already reflects everything incl. fundamentals (in line with EMH).*

→ ***Price moves in trends****: more likely to continue the past trend.*

→ ***History tends to repeat itself****: chart patterns repeat.*

2.4.1 The Dow Theory TYK T2

→ *Charles Dow (founder of Dow Jones), late 19th century; measures relative strength of the market;* ***barometer of business****.*

→ *Based on two indices:* ***DJIA*** *(Industrial) and* ***DJTA*** *(Transportation) — aggregate impact of all information.*

→ *Three simultaneous movements:*

  → ***Primary*** *— main trend (bull/bear),* ***1 year to 36 months*** *or longer.*

  → ***Secondary*** *— shorter,* ***opposite*** *direction,* ***2 weeks to a month*** *or more.*

  → ***Daily fluctuations*** *— not part of Dow interpretation, but study them as they make up longer movements.*

→ *Tells where the market is and where it is going,* ***not how far****. Successive* ***higher highs and higher lows*** *→* ***bullish****; lower highs and lower lows → bearish.*

→ *Primary uptrend:* ***three moves up*** *— (1) accumulation by far-sighted investors; (2) first reports of good earnings; (3) widespread reports of well-being + rampant speculation. Then far-sighted sell → first move down; non-sustainability confirmed → second;* ***distress selling*** *→ third.*

2.4.2 Elliot Wave Theory

Asked: May 2026 Q4(c) 4m

TYK T3

- → **Ralph Elliot, 1934**; inspired by Dow Theory and nature; based on **75 years** of price movements and charts.
- → **Wave** = movement of market price from one change in direction to the next change in the same direction; caused by buying/selling impulses from demand-supply pressures.
- → **Impulsive patterns (basic waves)**: **3 or 5 waves** in the direction of the basic movement (bull or bear phase).
- → **Corrective patterns (reaction waves)**: **3 waves** against the basic direction; correct the earlier rise (bull) or fall (bear).
- → Waves 1, 3, 5 = directional; waves 2, 4 = corrective.
- → **Complete cycle**: five-wave impulse + three-wave correction (a, b, c) = **8 waves**; two phases, bullish and bearish; then a fresh cycle starts.

2.4.3 Random Walk Theory

- → Dow-type theories assume trends give clues to future prices (technical analysis — much subjective judgment).
- → **Random walk hypothesis**: prices are unpredictable; **no relationship** between present and future prices; prices are independent — mere statistical ups and downs.
- → **M. G. Kendell** (British statistician): price changes behave as if generated by a **roulette wheel**, each outcome independent of past. Layman: a drunk walking in a blind lane.
- → Argument: (a) prices can never be predicted; (b) trends are not due to underlying factors but a statistical expression of past data; (c) no connection between successive **peaks and troughs**.

2.5 Charting Techniques

- → **Line chart**: closing price of each period (day/week/month) plotted and joined by a line.
- → **Bar chart**: vertical bar from low to high; short horizontal ticks for open and close.
- → **Japanese candlestick**: same OHLC data; body colour — **white** = close > open (**bullish**), **black** = close < open (**bearish**); open ≈ close = **Doji** (simple bar). Vertical line = high-low; rectangle = open-close.
- → **Point and figure**: more complex; detects **reversals**. Decide **box size** (e.g. ₹ 1, 2, 0.50 — smaller = more sensitive) and **reversal criterion** (boxes to be retraced to start a new column in the opposite direction). X = rising, O = falling.

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SM example (A Ltd., 6 days, O/H/L/C): 01-Jan 58/72/52/68; 02-Jan 71/73/58/64.30; 03-Jan 66/67/56/57; 04-Jan 58.50/75.50/55/72; 05-Jan 73.50/75/58/71; 06-Jan 74.50/76/55/74.50 — plotted as bar and candlestick charts. **P&F example:** prices 24, 26, 27, 26, 28, 27, 26, 25, 26, 23 over 10 periods — a column of X then a column of O on a 22–30 grid.

App-written reading of the SM data: 01 and 04 Jan white; 02, 03, 05 Jan black; 06 Jan (open = close 74.50) Doji.

2.6 Market Indicators

- → **Breadth index**: covers all securities traded = **net advances or declines ÷ number of issues traded**. Supports Dow Jones averages → technical **strength**; does not → **weakness** (market to move opposite). Addition to Dow Theory.
- → **Volume of transactions**: rising index with **increasing volume** → **buy** (unsatisfied demand); falling with increasing volume → bear, prices fall further; rising with decreasing volume → bull; falling with dwindling volume → bear. Best used with another indicator (Dow).
- → **Confidence index**: willingness to take a chance = **high-grade bond yields ÷ low-grade bond yields**. Rising index precedes rising market; fall precedes a drop (low-grade yields rising faster / falling slower). Usually, not always, **leading** → use with other indicators.
- → **Relative strength analysis**: some securities rise faster in bull / fall slower in bear; strength persists → invest in them. Measured by historical average returns, or ratios vs industry / vs market.
- → **Odd-lot theory**: **contrary-opinion** — the average person is usually wrong; used mainly to predict **tops in bull markets** and reversals in individual securities.

2.7 Support and Resistance Levels

- → Fall from a peak → peak = **resistance**; rebound from a trough → lowest value = **support**. Price moves between them.
- → Near resistance → **selling pressure** (those who missed selling at the high); near support → **buying pressure**. A **breach** = departure from status quo, new levels.

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SM example (A and B, Dec 2005 – Feb 2006 prices): A — support about **₹ 100**, resistance **₹ 125**; B — support **₹ 140**, resistance **₹ 160**.

2.8 Interpreting Price Patterns

Asked: Jan 2026 Q4(c) 4m

Pattern	Shape	Signal
(a) Channel	Series of uniformly changing tops and bottoms	Upward slope → rising prices; downward → declining
(b) Wedge	Tops and bottoms change in **opposite** directions, or same direction at **different rates**	—
(c) Head and shoulders	Large hump (head) between two smaller humps (shoulders); **neckline** joins where head meets shoulders. Price after 2nd shoulder is crucial.	Most important **reversal** pattern. Below neckline → drop ≈ distance from top of head to neckline.
(i) H&S top	Left shoulder, head, right shoulder	**Bearish**; fall below neckline → **sell**
(ii) Inverse H&S	Inverse of the top formation	**Bullish**; rise above neckline → **buy**
(d) Triangle / coil	—	**Uncertainty**; hard to predict breakout direction
(e) Flags and pennants	—	Previous trend likely to **continue**
(f) Double top	—	**Bearish**, price expected to fall
(g) Double bottom	—	**Bullish**, price expected to rise
(h) Gap	Opening price today – previous close	Wider gap → stronger signal of **continuation**; up-gap in rising market = eager buyers; lower gap in falling market = extreme selling pressure

2.9 Decision Using Data Analysis (Moving Averages)

Formula	Terms / logic	SM figures
$AMA_{n,t} = 1/n\ [P_t + P_{t-1} + \ldots + P_{t-(n-1)}]$	Simple average of the last n prices; equal weight 1/n each.	Trend: **200-day** (or 30-week) = long term; **60-day** = intermediate; **10-day** = short term.
$EMA_t = aP_t + (1 - a)\ EMA_{t-1}$; = (Close today − EMA_{t-1}) × a + EMA_{t-1}; a = 2 ÷ (n + 1)	a = exponent / smoothing constant; P_t = today's price; EMA_{t-1} = previous day's EMA; n = days. Decreasing weights, **highest on latest price**.	TYK PI: a = 0.062 (31-day)

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SM example (5-day MA, 10 days' closes 25.00, 26.00, 25.50, 24.50, 26.00, 26.00, 26.50, 26.50, 26.00, 27.00): 5-day sums 127.00, 128.00, 128.50, 129.50, 131.00, 132.00 → two-item centred totals 255.00, 256.50, 258.00, 260.50, 263.00 → MA (÷ 10) **25.50, 25.65, 25.80, 26.05, 26.30**.

Buy signal	Sell signal
(a) Price rises through MA line when MA is flattening out	(a) Price falls through MA line when MA is flattening out
(b) Price falls below MA line which is **rising**	(b) Price rises above MA line which is **falling**
(c) Price above MA falls but rises again before reaching MA	(c) Price below MA rises but falls again before reaching MA

Note (app-written): SM sell (c) prints "which is slow moving average line"; read as "below" (mirror of buy (c)).

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TYK PI (Sensex, EMA, previous EMA 15,000, a 0.062): 6th 14522 → diff (478) × 0.062 = (29.636) → **14970.364**; 7th 14925 → **14967.55**; 10th 15222 → **14983.32**; 11th 16000 → **15046.354**; 12th 16400 → **15130.28**; 13th 17000 → **15246.202**; 17th 18000 → diff 2753.798 × 0.062 = 170.735 → **15416.937**. Non-trading days skipped. Conclusion: **bullish** short to medium term → take **long** position.

Note (app-written): TYK PI question says "NSE Nifty" but the table and answer use Sensex; and 2 ÷ (31+1) = 0.0625, ICAI uses 0.062 as given.

2.10 Evaluation of Technical Analysis

Asked: Sep 2025 Q4(c) 4m

Arguments in favour	Arguments against
→ **Crowd psychology** makes trends persist; tools spot them early.	→ No **convincing explanation** for tools used.
→ Shifts in demand and supply are **gradual**, not instantaneous; detected early.	→ Empirical evidence for **random walk** casts a shadow.
→ Fundamental information is **assimilated over time**; price moves in the same direction till fully assimilated.	→ By the time a trend is signalled it may **already have taken place**.
	→ **Self-defeating**: more users → less value.

Nutshell: may not work in a rational, efficient market; helpful in real-world imperfect markets; best used **with fundamental analysis**.

3 Differences between Fundamental and Technical Analysis

Basis	Fundamental	Technical
Method	Macro factors (GDP, inflation, interest rate, growth rate) + company micro factors (sales, profitability, solvency, assets & liabilities, cash)	Predicts prices and direction from purely **historical market data** — price movements, volume, open interest
Rule	Prices of a share discounts everything	Price captures everything
Usefulness	**Long-term** investing	**Short-term** investing

4 Efficient Market Theory (EMH) TYK TI

→ **Eugene Fama** (University of Chicago), 1960s. All available price-sensitive information is **fully reflected** in prices → no one can consistently **beat the market**; stocks trade at fair value; can't buy undervalued / sell overvalued.

→ Expert stock selection or market timing can't outperform; higher return only by buying **riskier investments**.

4.1 Search for Theory

Randomness of prices is the result of an efficient market:

→ Information **freely and instantaneously** available to all.

→ **Keen competition** → prices reflect intrinsic values (impound all information).

→ Prices change only in response to **new information**, unrelated to previous → unpredictable.

4.2 Misconception about Efficient Market Theory

→ EMT implies prices factor in all information, so no consistent long-term returns from market operations.

→ Prices fluctuate and cannot reflect fair value, as the **future is uncertain** — surprises keep coming.

→ Failure of institutional portfolio managers to outperform is taken to mean **lack of competence**; in fact efficiency exists **because** managers do the job well in a competitive setting.

→ Random movement is taken to mean the market is **irrational**; but **randomness ≠ irrationality** — rational, competitive investors make price changes random.

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TYK TI (EMT and its major misconception) → ICAI answer: refer para 4.

4.3 Level of Market Efficiency

Form (Fama)	Price reflects
Weak	All information in the record of **past prices and volumes**
Semi-strong	Past prices and volumes + all other **publicly available** information
Strong	All information, **public and private** (highest order)

4.4 Empirical Evidence on Weak Form

No relationship between past and future price movements. Three tests:

→ **(a) Serial correlation test**: correlate price change in one period with another; studies (different stocks, lags, periods — daily, weekly, fortnightly, monthly) found **no significant serial correlation**. Not long-term trends, where prices tend to move upwards.

→ **(b) Run test**: mark each change + (increase) or − (decrease). A **run** continues while the sign is the same; a change of sign ends it. Compare number of runs with a purely random series of the same size. Results strongly support the **random walk** model.

→ **(c) Filter rules test**: buy when price rises at least N%, hold till it falls at least N% from a subsequent high, then sell. Studies: filter rules do **not outperform buy and hold**, esp. after commissions.

Run test formula (TYK P2 answer)	Terms / logic
$\mu_r = [2n_1n_2 \div (n_1 + n_2)] + 1$	Expected runs in a random series; n_1 = no. of + changes, n_2 = no. of − changes; r = actual runs.
$\sigma_r = \sqrt{\{2n_1n_2(2n_1n_2 - n_1 - n_2) \div [(n_1 + n_2)^2(n_1 + n_2 - 1)]\}}$	Standard deviation of runs.
Limits = $\mu_r \pm t \times \sigma_r$	Two-tailed (too few runs → not random); t at n − 1 df (n = no. of changes). r within limits → random → **weak-form efficient**.

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TYK P2 (index, Oct 2007, 20 closes → 19 changes): runs r = **8**; n_1 = 11, n_2 = 8; μ = 176/19 + 1 = **10.26**; $\sigma = \sqrt{[176 \times 157 \div (19^2 \times 18)]} = \sqrt{4.252}$ = **2.06**. 5% (t 2.101, 18 df): **5.932 to 14.588**; 10% (t 1.734): **6.688 to 13.832**. r = 8 within both → market is **weak-form efficient**.

4.5 Empirical Evidence on Semi-strong Form

- Prices adjust **rapidly** to all public information → no above-normal risk-adjusted return from public information. Tests: can public information earn above-normal returns, and how fast do prices adjust to earnings, dividends, mergers, acquisitions, stock splits?
- **Fama, Fisher, Jensen and Roll**: **940 stock splits**, NYSE, 1957–1959 — prior to the split, stock earns higher returns than predicted by any market model.
- **Ball and Brown** (SM prints "Boll"): annual earnings announcements; group 1 earnings up vs average, group 2 down. SM: before the announcement group 1 earned **positive abnormal** returns; group 2 earned **negative abnormal** returns; stock in both groups earned **normal** returns (after the announcement).
- **Documented inefficiencies / anomalies**: prices adjust **gradually** to unanticipated quarterly earnings; **small firms** outperform large; **low P/E** outperforms high P/E; **Monday's** return lower than other days.

Note (app-written): the SM sentence reads "stock in the second group earned negative abnormal returns after the announcement of earnings. Stock in both the groups earned normal returns" — the wording is muddled; read it as abnormal returns before the announcement and normal returns after it.

4.6 Empirical Evidence on Strong Form

- All information, public or private, reflected — an **extreme** hypothesis.
- Tested on returns of corporate insiders, stock exchange specialists, mutual fund managers. **Insiders** (internal information) and **specialists** (monopolistic exposure) earn **superior** risk-adjusted returns.
- **Mutual fund managers** do not on average; professionally managed portfolios no better than random portfolios — **Burton Malkiel**, *A Random Walk Down Wall Street*.

4.7 Challenges to the Efficient Market Theory

- **Information inadequacy**: not freely available or rapidly transmitted; companies circulate misinformation.
- **Limited information processing**: **Herbert Simon** — millions of bits per second generated, perception admits not more than a thousand; **David Dreman** — under anxiety and uncertainty the market can become a giant.
- **Irrational behaviour**: **J. M. Keynes** — considerations irrelevant to prospective yield enter valuation; **L. C. Gupta** — market works haphazardly, like a blind man firing a gun; hit or miss tactics.
- **Monopolistic influence**: powerful institutions and big operators influence prices → less competitive.

5 Equity Research and Tools Available

- Area of finance / investment banking analysing a company's financial performance and other factors to decide **buy, sell or hold**; also used in M&A to decide the **swap / exchange ratio**. Done by Equity Research Analysts; earlier paper based, now online tools.

Tool	Founded	Tool	Founded
Bloomberg Terminal	1981	Koyfin	2016
Benzinga Pro	2010	Finbox.io	2014
Refinitiv EIKON	2010	GuruFocus	2004
MarketXLS	2015	Business Quant	2014
Stockopedia	2010	Ycharts	2009

Exam steps — EMA: $a = 2 \div (n+1)$ (or as given) → skip non-trading days → diff = close − previous EMA → × a → add to previous EMA → carry forward → rising EMA = bullish, take long position (TYK PI).

Exam steps — Run test: sign each change (+/−) → count runs r, n_1, n_2 → $\mu_r = 2n_1n_2/(n_1+n_2) + 1$ → σ_r → limits $\mu \pm t\,\sigma$ (df = changes − 1) → r inside = weak-form efficient.

Exam steps — Moving average: sum of n most recent closes → (SM) two-item centred total of adjacent sums → $\div 2n$ (SM: ÷ 10 for 5-day) → compare price line with MA line using buy/sell signals.

Exam steps — Intrinsic value: $D_1 = D_0(1+g)$ or bE_1 → check $g < k$ → $P_0 = D_1/(k-g)$ → price < value buy, > value sell.

④ Security Analysis — what ICAI asked (Sep 2025, Jan 2026, May 2026)

12 marks *set across the last three attempts (3.4% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	*Q4(c) [either]*	*4*	*Technical analysis: favourable and unfavourable arguments (Theory)*
Jan 2026	*Q4(c)*	*4*	*Technical analysis: identify chart patterns (uptrend, continuation/flag, head and shoulders, triangle/wedge) from descriptions, any four (Theory)*
May 2026	*Q4(c) [either]*	*4*	*Elliott Wave Theory of technical analysis (Theory)*

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain the Efficient Market Theory in and what is major misconception about this theory?

ICAI answer: "Please refer paragraph 4" – SM paragraph 4 reproduced

4. EFFICIENT MARKET THEORY (EFFICIENT MARKET HYPOTHESIS)

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Stating otherwise this theory states that no one can "beat the market" hence making it impossible for investors to either purchase undervalued stocks or sell stocks for inflated prices as stocks are always traded at their fair value on stock exchanges. Hence it is impossible to outperform the overall market through expert stock selection or market timing and that the only way an investor can possibly obtain higher returns is by purchasing risky investments.

4.1 Search for Theory

When empirical evidence in favour of Random walk hypothesis seemed overwhelming, researchers wanted to know about the Economic processes that produced a Random walk. They concluded that randomness of stock price was a result of efficient market that led to the following view points:

- Information is freely and instantaneously available to all market participants.
- Keen competition among the market participants more or less ensures that market will reflect intrinsic values. This means that they will fully impound all available information.
- Price change only response to new information that is unrelated to previous information and therefore unpredictable.

4.2 Misconception about Efficient Market Theory

Efficient Market Theory implies that market prices factor in all available information and as such it is not possible for any investor to earn consistent long term returns from market operations.

Although price tends to fluctuate they cannot reflect fair value. This is because the future is uncertain. The market springs surprises continually and as prices reflect the surprises they fluctuate.

Inability of institutional portfolio managers to achieve superior investment performance implies that they lack competence in an efficient market. It is not possible to achieve superior investment performance since market efficiency exists due to portfolio mangers doing this job well in a competitive setting.

The random movement of stock prices suggests that stock market is irrational. Randomness and irrationality are two different things, if investors are rational and competitive, price changes are bound to be random.

4.3 Level of Market Efficiency

That price reflects all available information, the highest order of market efficiency. According to Eugene Fama, there exist three levels of market efficiency:-

(i) *Weak form efficiency* – Price reflects all information found in the record of past prices and volumes.

(ii) *Semi – Strong efficiency* – Price reflects not only all information found in the record of past prices and volumes but also all other publicly available information.

(iii) *Strong form efficiency* – Price reflects all available information public as well as private.

4.4 Empirical Evidence on Weak form of Efficient Market Theory

According to the Weak form Efficient Market Theory current price of a stock reflect all information found in the record of past prices and volumes. This means that there is no relationship between the past and future price movements.

Three types of tests can be employed to empirically verify the weak form of Efficient Market Theory- Serial Correlation Test, Run Test and Filter Rule Test.

(a) *Serial Correlation Test*: To test for randomness in stock price changes, one has to look at serial correlation. For this purpose, price change in one period has to be correlated with price change in some other period. Price changes are considered to be serially independent. Serial correlation studies employing different stocks, different time lags and different time period have been conducted to detect serial correlation but no significant serial correlation could be discovered. These studies were carried on short term trends viz. daily, weekly, fortnightly and monthly and not in long term trends in stock prices as in such cases. Stock prices tend to move upwards.

(b) *Run Test*: Given a series of stock price changes each price change is designated + if it represents an increase and – if it represents a decrease. The resulting series may be -,+, - , -, - , +, +.

A run occurs when there is no difference between the sign of two changes. When the sign of change differs, the run ends and new run begins.

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To test a series of price change for independence, the number of runs in that series is compared with a number of runs in a purely random series of the size and in the process determines whether it is statistically different. By and large, the result of these studies strongly supports the Random Walk Model.

(c) *Filter Rules Test*: If the price of stock increases by at least N% buy and hold it until its price decreases by at least N% from a subsequent high. When the price decreases at least N% or more, sell it. If the behaviour of stock price changes is random, filter rules should not apply in such a buy and hold strategy. By and large, studies suggest that filter rules do not out perform a single buy and hold strategy particular after considering commission on transaction.

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number of studies was conducted which lead to the following queries: Whether it was possible to earn on the above normal rate of return after adjustment for risk, using only publicly available information and how rapidly prices adjust to public announcement with regard to earnings, dividends, mergers, acquisitions, stock-splits?

Several studies support the Semi-strong form Efficient Market Theory. Fama, Fisher, Jensen and Roll in their adjustment of stock prices to new information examined the effect of stock split on return of 940 stock splits in New York Stock Exchange during the period 1957-1959 They found that prior to the split, stock earns higher returns than predicted by any market model.

Boll and Brown in an empirical evaluation of accounting income numbers studied the effect of annual earnings announcements. They divided the firms into two groups. First group consists of firms whose earnings increased in relation to the average corporate earnings while second group consists of firms whose earnings decreased in relation to the average corporate earnings. They found that before the announcement of earnings, stock in the first group earned positive abnormal returns while stock in the second group earned negative abnormal returns after the announcement of earnings. Stock in both the groups earned normal returns.

There have been studies which have been empirically documented showing the following inefficiencies and anomalies:

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According to the Efficient Market Theory, all available information, public or private, is reflected in the stock prices. This represents an extreme hypothesis.

To test this theory, the researcher analysed returns earned by certain groups viz. corporate insiders, specialists on stock exchanges, mutual fund managers who have access to internal information (not publicly available), or possess greater resource or ability to intensively analyse information in the public domain. They suggested that corporate insiders (having access to internal information) and stock exchange specialists (having monopolistic exposure) earn superior rate of return after adjustment of risk.

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Theory Q2

2. Explain Dow theory.

ICAI answer: "Please refer paragraph 2.4.1" – SM paragraph 2.4.1 reproduced

2.4.1 The Dow Theory

The Dow Theory is one of the oldest and most famous technical theories. It was originated by Charles Dow, the founder of Dow Jones Company in late nineteenth century. It is a helpful tool for determining the relative strength of the stock market. It can also be used as a barometer of business.

The Dow Theory is based upon the movements of two indices, constructed by Charles Dow, Dow Jones Industrial Average (DJIA) and Dow Jones Transportation Average (DJTA). These averages reflect the aggregate impact of all kinds of information on the market. The movements of the market are divided into three classifications, all going at the same time; the primary movement, the secondary movement, and the daily fluctuations. The primary movement is the main trend of the market, which lasts from one year to 36 months or longer. This trend is commonly called bear or bull market. The secondary movement of the market is shorter in duration than the primary movement, and is opposite in direction. It lasts from two weeks to a month or more. The daily fluctuations are the narrow movements from day-to-day. These fluctuations are not part of the Dow Theory interpretation of the stock market. However, daily movements must be carefully studied, along with primary and secondary movements, as they go to make up the longer movement in the market.

Thus, the Dow Theory's purpose is to determine where the market is and where is it going, although not how far or high. The theory, in practice, states that if the cyclical swings of the stock market averages are successively higher and the successive lows are higher, then the market trend is up and a bullish market exists. Contrarily, if the successive highs and successive lows are lower, then the direction of the market is down and a bearish market exists.

Charles Dow proposed that the primary uptrend would have three moves up, the *first* one being caused by accumulation of shares by the far-sighted, knowledgeable investors, the *second* move would be caused by the arrival of the first reports of good earnings by corporations, and the last move up would be caused by widespread report of financial well-being of corporations. The *third* stage would also see rampant speculation in the market. Towards the end of the third stage, the far-sighted investors, realizing that the high earnings levels may not be sustained, would start selling, starting the first move down of a downtrend, and as the non-sustainability of high earnings is

confirmed, the second move down would be initiated and then the third move down would result from distress selling in the market.

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Theory Q3

3. Explain the Elliot Wave Theory of technical analysis.

ICAI answer: "Please refer paragraph 2.4.2" – SM paragraph 2.4.2 reproduced

2.4.2 Elliot Wave Theory

Inspired by the Dow Theory and by observations found throughout nature, Ralph Elliot formulated Elliot Wave Theory in 1934. This theory was based on analysis of 75 years stock price movements and charts. From his studies, he defined price movements in terms of waves. Accordingly, this theory was named Elliot Wave Theory. Elliot found that the markets exhibited certain repeated patterns or waves. As per this theory wave is a movement of the market price from one change in the direction to the next change in the same direction. These waves are resulted from buying and selling impulses emerging from the demand and supply pressures on the market. Depending on the demand and supply pressures, waves are generated in the prices.

As per this theory, waves can be classified into two parts:-

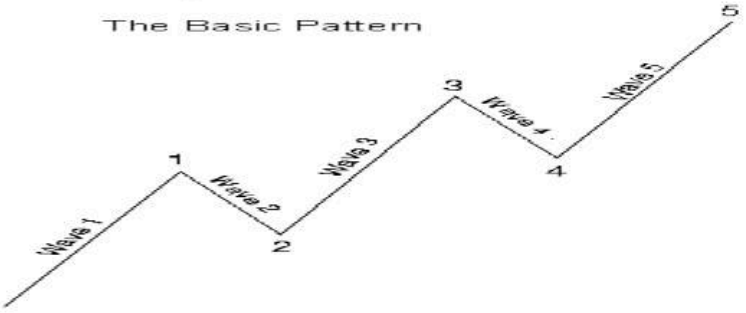
- Impulsive patterns
- Corrective patters

Let us discuss each of these patterns.

(a) **Impulsive Patterns-(Basic Waves) -** In this pattern there will be 3 or 5 waves in a given direction (going upward or downward). These waves shall move in the direction of the basic movement. This movement can indicate bull phase or bear phase.

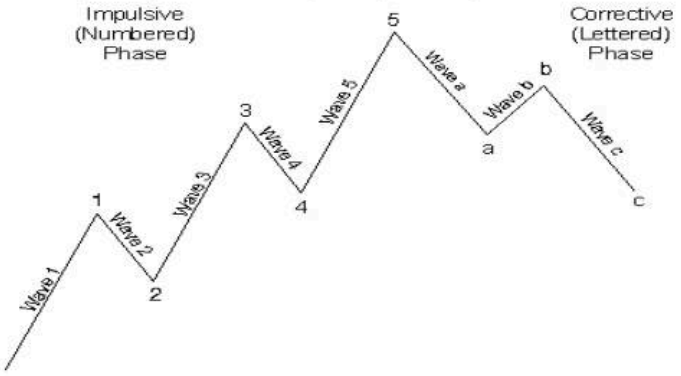
(b) **Corrective Patterns- (Reaction Waves) -** These 3 waves are against the basic direction of the basic movement. Correction involves correcting the earlier rise in case of bull market and fall in case of bear market.

As shown in the following diagram waves 1, 3 and 5 are directional movements, which are separated or corrected by wave 2 & 4, termed as corrective movements.



Source: http://elliotwave.net/

Complete Cycle -As shown in following figure five-wave impulses is following by a three-wave correction (a,b & c) to form a complete cycle of eight waves.



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One complete cycle consists of waves made up of two distinct phases, bullish and bearish. On completion of full one cycle i.e. termination of 8 waves movement, the fresh cycle starts with similar impulses arising out of market trading.

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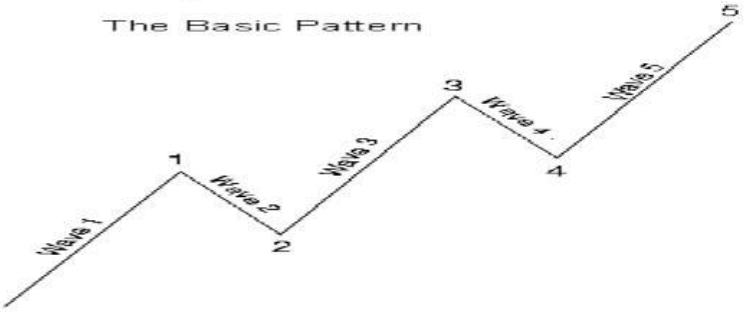
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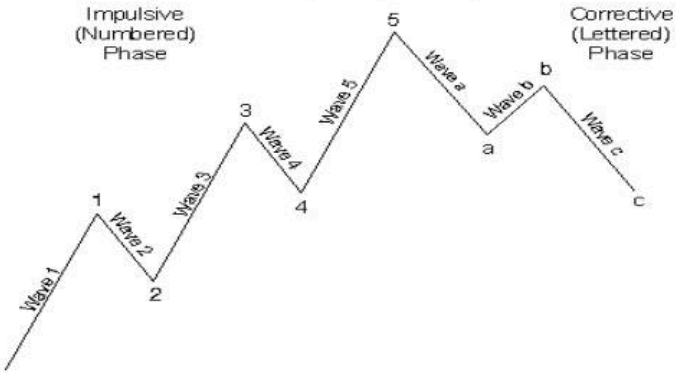
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Theory Q4

4. Explain the various indicators that can be used to assess the performance of an economy.

ICAI answer: "Please refer paragraph 1.2.2" – SM paragraph 1.2.2 reproduced

1.2.2 Techniques Used in Economic Analysis

Economic analysis is used to forecast national income with its various components that have a bearing on the concerned industry and the company in particular. Gross National Product (GNP) is used to measure national income as it reflects the growth rate in economic activities and has been regarded as a forecasting tool for analyzing the overall economy along with its various components during a particular period.

Some of the techniques used for economic analysis are:

(a) Anticipatory Surveys: They help investors to form an opinion about the future state of the economy. It incorporates expert opinion on construction activities, expenditure on plant and machinery, levels of inventory – all having a definite bearing on economic activities. Also future spending habits of consumers are taken into account.

In spite of valuable inputs available through this method, it has certain drawbacks:

(i) Survey results do not guarantee that intentions surveyed would materialize.

(ii) They are not regarded as forecasts per se, as there can be a consensus approach by the investor for exercising his opinion.

Continuous monitoring of this practice is called for to make this technique popular.

(b) Barometer/Indicator Approach: Various indicators are used to find out how the economy shall perform in the future. The indicators have been classified as under:

(i) *Leading Indicators*: They lead the economic activity in terms of their outcome. They relate to the time series data of the variables that reach high/low points in advance of economic activity.

(ii) *Roughly Coincidental Indicators:* They reach their peaks and troughs at approximately the same in the economy.

(iii) *Lagging Indicators:* They are time series data of variables that lag behind in their consequences vis-a- vis the economy. They reach their turning points after the economy has reached its own already.

All these approaches suggest direction of change in the aggregate economic activity but nothing about its magnitude. The various measures obtained form such indicators may give conflicting

signals about the future direction of the economy. To avoid this limitation, use of diffusion/composite index is suggested whereby combining several indicators into one index to measure the strength/weaknesses in the movement of a particular set of indicators. Computation of diffusion indices is no doubt difficult notwithstanding the fact it does not eliminate irregular movements.

Money supply in the economy also affects investment decisions. Rate of change in money supply in the economy affects GNP, corporate profits, interest rates and stock prices. Increase in money supply fuels inflation. As investment in stocks is considered as a hedge against inflation, stock prices go up during inflationary period.

(c) Economic Model Building Approach*:* In this approach, a precise and clear relationship between dependent and independent variables is determined. GNP model building or sectoral analysis is used in practice through the use of national accounting framework. The steps used are as follows:

(i) Hypothesize total economic demand by measuring total income (GNP) based on political stability, rate of inflation, changes in economic levels.

(ii) Forecasting the GNP by estimating levels of various components viz. consumption expenditure, gross private domestic investment, government purchases of goods/services, net exports.

(iii) After forecasting individual components of GNP, add them up to obtain the forecasted GNP.

(iv) Comparison is made of total GNP thus arrived at with that from an independent agency for the forecast of GNP and then the overall forecast is tested for consistency. This is carried out for ensuring that both the total forecast and the component wise forecast fit together in a reasonable manner.

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(ii) *Roughly Coincidental Indicators:* They reach their peaks and troughs at approximately the same in the economy.

(iii) *Lagging Indicators:* They are time series data of variables that lag behind in their consequences vis-a- vis the economy. They reach their turning points after the economy has reached its own already.

All these approaches suggest direction of change in the aggregate economic activity but nothing about its magnitude. The various measures obtained form such indicators may give conflicting

signals about the future direction of the economy. To avoid this limitation, use of diffusion/composite index is suggested whereby combining several indicators into one index to measure the strength/weaknesses in the movement of a particular set of indicators. Computation of diffusion indices is no doubt difficult notwithstanding the fact it does not eliminate irregular movements.

Money supply in the economy also affects investment decisions. Rate of change in money supply in the economy affects GNP, corporate profits, interest rates and stock prices. Increase in money supply fuels inflation. As investment in stocks is considered as a hedge against inflation, stock prices go up during inflationary period.

(c) Economic Model Building Approach*:* In this approach, a precise and clear relationship between dependent and independent variables is determined. GNP model building or sectoral analysis is used in practice through the use of national accounting framework. The steps used are as follows:

(i) Hypothesize total economic demand by measuring total income (GNP) based on political stability, rate of inflation, changes in economic levels.

(ii) Forecasting the GNP by estimating levels of various components viz. consumption expenditure, gross private domestic investment, government purchases of goods/services, net exports.

(iii) After forecasting individual components of GNP, add them up to obtain the forecasted GNP.

(iv) Comparison is made of total GNP thus arrived at with that from an independent agency for the forecast of GNP and then the overall forecast is tested for consistency. This is carried out for ensuring that both the total forecast and the component wise forecast fit together in a reasonable manner.

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. Closing values of NSE Nifty from 6th to 17th day of the month of January of the year 2020 were as follows:

Days	Date	Day	Sensex
1	6	THU	14522
2	7	FRI	14925
3	8	SAT	No Trading
4	9	SUN	No Trading
5	10	MON	15222
6	11	TUE	16000
7	12	WED	16400
8	13	THU	17000
9	14	FRI	No Trading
10	15	SAT	No Trading
11	16	SUN	No Trading
12	17	MON	18000

Calculate Exponential Moving Average (EMA) of Sensex during the above period. The previous day exponential moving average of Sensex can be assumed as 15,000. The value of exponent for 31 days EMA is 0.062.

Give detailed analysis on the basis of your calculations.

Solution — ICAI

1.

Date	1 Sensex	2 EMA for Previous day	3 1-2	4 3×0.062	5 EMA 2 ± 4
6	14522	15000	(478)	(29.636)	14970.364
7	14925	14970.364	(45.364)	(2.812)	14967.55
10	15222	14967.55	254.45	15.776	14983.32
11	16000	14983.32	1016.68	63.034	15046.354
12	16400	15046.354	1353.646	83.926	15130.28
13	17000	15130.28	1869.72	115.922	15246.202
17	18000	15246.202	2753.798	170.735	15416.937

Conclusion – The market is bullish. The market is likely to remain bullish for short term to medium term if other factors remain the same. On the basis of this indicator (EMA) the investors/brokers can take long position.

Practical Q2

2. The closing value of a Stock Market Index for the month of October, 2007 is given below:

Date Closing	Index Value
1.10.07	2800
3.10.07	2780
4.10.07	2795
5.10.07	2830
8.10.07	2760
9.10.07	2790
10.10.07	2880
11.10.07	2960
12.10.07	2990
15.10.07	3200
16.10.07	3300
17.10.07	3450
19.10.07	3360
22.10.07	3290
23.10.07	3360
24.10.07	3340
25.10.07	3290
29.10.07	3240
30.10.07	3140
31.10.07	3260

You are required to test the weak form of efficient market hypothesis by applying the run test at 5% and 10% level of significance.

Following values can be used:

Value of t at 5% is 2.101 at 18 degrees of freedom

Value of t at 10% is 1.734 at 18 degrees of freedom

Solution — ICAI

2.

Date	Closing Index	Sign of Price Charge
1.10.07	2800	
3.10.07	2780	-
4.10.07	2795	+
5.10.07	2830	+
8.10.07	2760	-
9.10.07	2790	+
10.10.07	2880	+
11.10.07	2960	+
12.10.07	2990	+
15.10.07	3200	+
16.10.07	3300	+
17.10.07	3450	+
19.10.07	3360	-
22.10.07	3290	-
23.10.07	3360	+
24.10.07	3340	-
25.10.07	3290	-
29.10.07	3240	-
30.10.07	3140	-
31.10.07	3260	+

Total of sign of price changes (r) = 8

No of Positive changes = $n_1 = 11$

No. of Negative changes = $n_2 = 8$

$$\mu_r = \frac{2n_1n_2}{n_1+n_2}+1$$

$$\mu = \frac{2\times11\times8}{11+8}+1 = 176/19 + 1 = 10.26$$

$$\hat{\sigma}_r = \sqrt{\frac{2n_1n_2(2n_1n_2-n_1-n_2)}{(n_1+n_2)^2(n_1+n_2-1)}}$$

$$\hat{\sigma}_r = \sqrt{\frac{(2\times11\times8)\,(2\times11\times8-11-8)}{(11+8)^2(11+8-1)}} = \sqrt{\frac{176\times157}{(19)^2(18)}} = \sqrt{4.252} = 2.06$$

Since too few runs in the case would indicate that the movement of prices is not random. We employ a two- tailed test the randomness of prices.

Test at 5% level of significance at 18 degrees of freedom using t- table

The lower limit

$= \mu - t \times \hat{\sigma}_r = 10.26 - 2.101 \times 2.06 = 5.932$

Upper limit

$= \mu + t \times \hat{\sigma}_r = 10.26 + 2.101 \times 2.06 = 14.588$

At 10% level of significance at 18 degrees of freedom

Lower limit

$= 10.26 - 1.734 \times 2.06 = 6.688$

Upper limit

$= 10.26 + 1.734 \times 2.06 = 13.832$

As seen r lies between these limits. Hence, the market exhibits weak form of efficiency.

*For a sample of size n, the t distribution will have n-1 degrees of freedom.

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